Course: Digital Logic Design

Number System

- Decimal
- Binary
- Hexa Decimal
- Octal
- Addition / Subtraction
- Multiplication / Division
- How Computer Add or Subtract

Gates

- Boolean Algebra
- types of Boolean Algebra
- K-Map
- etc

Circuits

- Combinational Circuit
- Can't store Data
- Sequential Circuit
- Can store Data

Number System

It is Basically a system in which we able to know how to express a Number and how to deal with the Number.



Add / Sub / Multiplication / Division



How to convert a number from one system to another



etc.

Number Systems

1. **Decimal Number**
   Base 10
2. **Binary Numbers**
   Base 2
3. **Octal Number**
   Base 8
4. **Hexadecimal Number**
   Base 16

Decimal Number System

Definition:

It is a standard number system used to represent integers and non-integer numbers.

This is the number system we use in daily life.

Digits:

0	1	2	3	4	5	6	7	8	9

Base (Radix): 10

Representation:

$(375)_{10}$ $(25.75)_{10}$ $(100)_{10}$ $(13.625)_{10}$

Binary Number System

Definition:

Binary is the language of computers. It uses only two digits.

It's used to store, access, and manipulate data in computers.

Digits:

0	1

Base (Radix): **2**

Representation:

$(101110111)_2$ $(11001)_2$ $(101101)_2$ $(1101.101)_2$

Octal Number System

Definition:

Octal number system uses 8 digits. One octal digit represents exactly 3 binary bits.

It is convenient for representing binary numbers in shorter form.

Digits:

0	1	2	3	4	5	6	7

Base (Radix): 8

Representation:

$(567)_8$ $(144)_8$ $(31.6)_8$ $(113.2)_8$

Hexadecimal Number System

Definition:

Hexadecimal uses 16 symbols. One hex digit represents exactly 4 binary bits.

Widely used in memory addresses, color codes, and debugging.

Digits & Characters:

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

A=10 B=11 C=12 D=13 E=14 F=15

Base (Radix): **16**

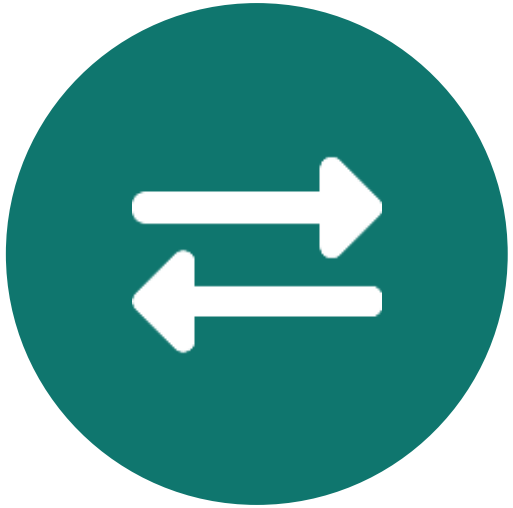
Representation:

$(177)_{16}$ $(FF)_{16}$ $(17D)_{16}$ $(2D.4)_{16}$

CONVERSIONS

Decimal → Binary / Octal / Hexadecimal

Binary / Octal / Hexadecimal → Decimal



DECIMAL TO BINARY CONVERSION

$(375)_{10}$ → Binary

2	375	
2	187	1
2	93	1
2	46	1
2	23	0
2	11	1
2	5	1
2	2	1
	1	0



Read remainders
bottom → top

Answer: $(101110111)_2$

DECIMAL TO BINARY CONVERSION

$(25)_{10}$ → Binary

2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

Read remainders
bottom → top

Answer: $(11001)_2$

DECIMAL TO BINARY CONVERSION

$(45)_{10}$ → Binary

2	45	
2	22	1
2	11	0
2	5	1
2	2	1
	1	0



Read remainders
bottom → top

Answer: $(101101)_2$

DECIMAL TO BINARY CONVERSION

$(100)_{10}$ → Binary

2	100	
2	50	0
2	25	0
2	12	1
2	6	0
2	3	0
	1	1



Read remainders
bottom → top

Answer: $(1100100)_2$

DECIMAL TO BINARY CONVERSION

$(13)_{10}$ → Binary

2	13	
2	6	1
2	3	0
2	1	1
	0	1

Read remainders
bottom → top

Answer: $(1101)_2$

DECIMAL TO BINARY (with Point)

$(75.25)_{10}$ → Binary

Integer Part

2	75	
2	37	1
2	18	1
2	9	0
2	4	1
2	2	0
	1	0

Fractional Part

0.25 × 2 = 0.50 → 0

0.50 × 2 = 1.00 → 1

top → bottom

Answer: $(1001011.01)_2$

DECIMAL TO BINARY (with Point)

$(13.625)_{10}$ → Binary

Integer: 13 → 1101_2

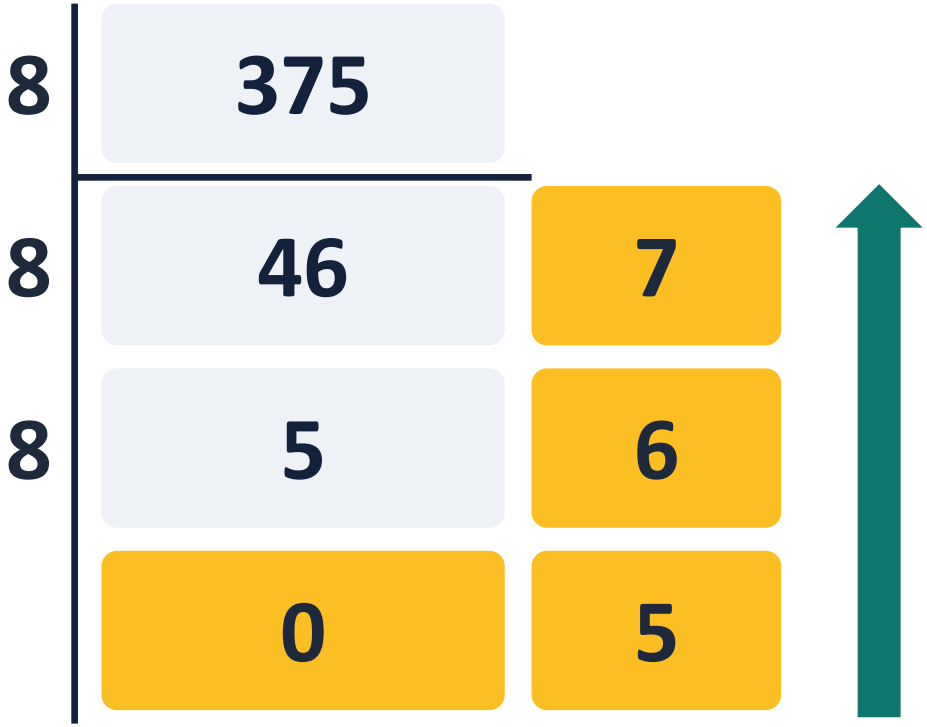
Fractional:

0.625 × 2 = 1.25	→	1
0.25 × 2 = 0.50	→	0
0.50 × 2 = 1.00	→	1

Answer: $(1101.101)_2$

DECIMAL TO OCTAL CONVERSION

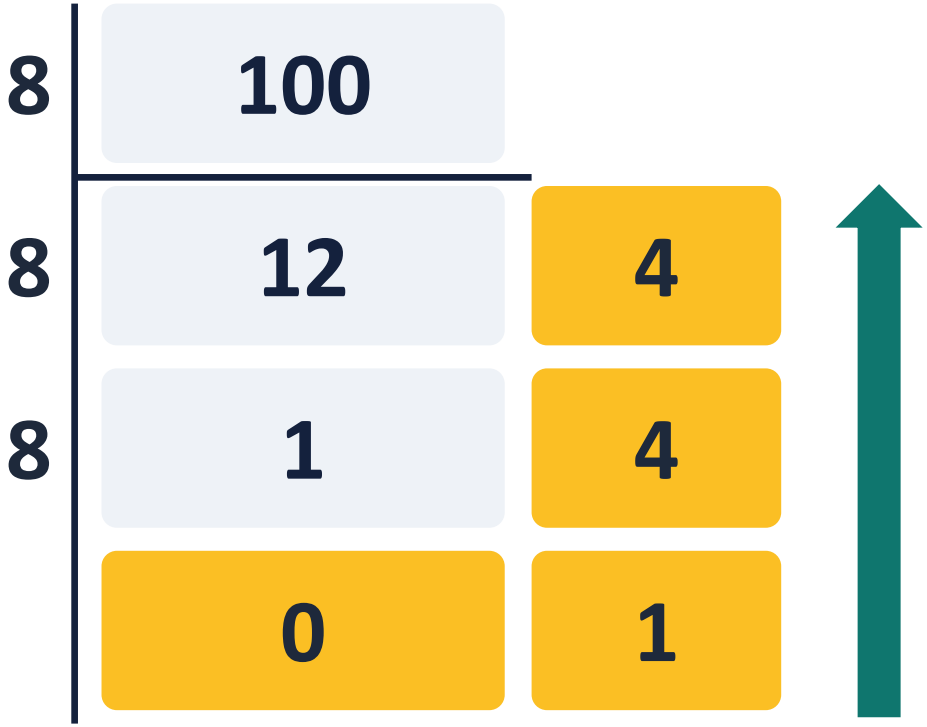
$(375)_{10}$ → Octal



Answer: $(567)_8$

DECIMAL TO OCTAL CONVERSION

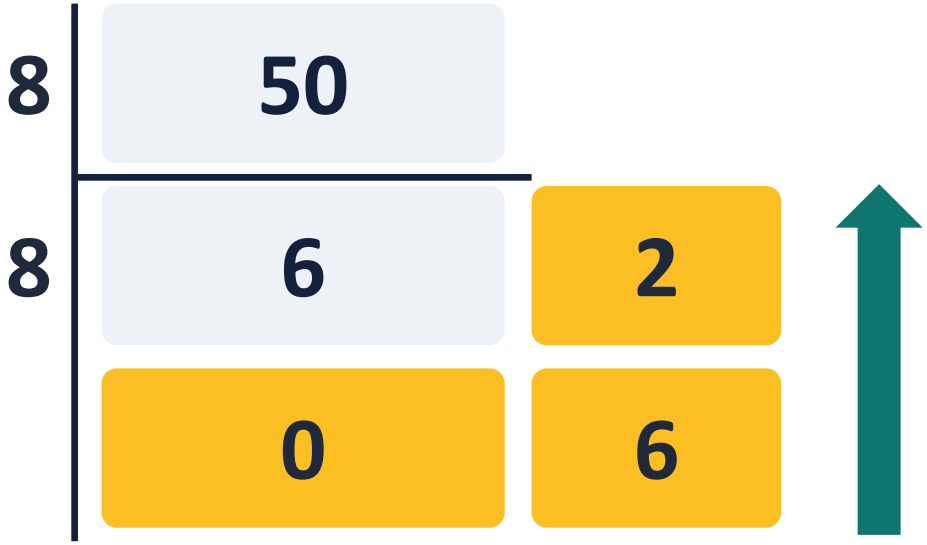
$(100)_{10}$ → Octal



Answer: $(144)_8$

DECIMAL TO OCTAL CONVERSION

$(50)_{10}$ → Octal



Answer: $(62)_8$

DECIMAL TO OCTAL (with Point)

$(75.25)_{10}$ → Octal

Integer Part

8	75	
8	9	3
	1	1



Fractional Part

0.25 × 8 = 2.00 → 2



top → bottom

Answer: $(113.2)_8$

DECIMAL TO OCTAL (with Point)

$(25.75)_{10}$ → Octal

Integer: 25 → 31_8

Fractional:

0.75 × 8 = 6.00 → **6**

Answer: $(31.6)_8$

DECIMAL TO HEXADECIMAL CONVERSION

$(381)_{10}$ → Hexadecimal

16	381	
16	23	13 (D)
	1	7



Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Answer: $(17D)_{16}$

DECIMAL TO HEXADECIMAL CONVERSION

$(255)_{10}$ → Hexadecimal

16	255	
16	15	15 (F)
	0	15 (F)



Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Answer: $(FF)_{16}$

DECIMAL TO HEXADECIMAL CONVERSION

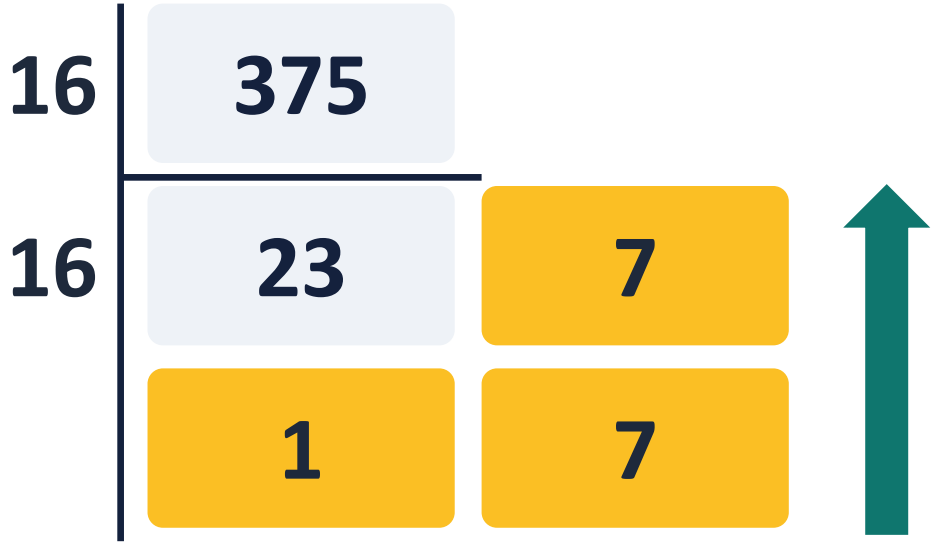
$(100)_{10}$ → Hexadecimal



Answer: $(64)_{16}$

DECIMAL TO HEXADECIMAL CONVERSION

$(375)_{10}$ → Hexadecimal



Answer: $(177)_{16}$

DECIMAL TO HEXADECIMAL (with Point)

$(76.25)_{10}$ → Hexadecimal

Integer: 76 → $4C_{16}$

16	76	
16	4	12 (C)
	0	4



Fractional:

0.25 × 16 = 4.00 → 4

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Answer: $(4C.4)_{16}$

DECIMAL TO HEXADECIMAL (with Point)

$(45.25)_{10}$ → Hexadecimal

Integer: 45 → $2D_{16}$

16	45	
16	2	13 (D)
	0	2



Fractional:

0.25 × 16 = 4.00 → 4

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Answer: $(2D.4)_{16}$

Binary → Decimal | $(1101.101)_2$

Position :	3	2	1	0	.	–1	–2	–3
Binary :	1	1	0	1	.	1	0	1

Value:

$1\times(2)^3 + 1\times(2)^2 + 0\times(2)^1 + 1\times(2)^0 + 1\times(2)^{-1} + 0\times(2)^{-2} + 1\times(2)^{-3}$

$$8 + 4 + 0 + 1 + 0.5 + 0 + 0.125 = (13.625)_{10}$$

Binary → Decimal | $(1011.11)_2$

Position :	3	2	1	0	.	–1	–2
Binary :	1	0	1	1	.	1	1

Value:

$$1\times(2)^3 + 0\times(2)^2 + 1\times(2)^1 + 1\times(2)^0 + 1\times(2)^{-1} + 1\times(2)^{-2}$$

$$8 + 0 + 2 + 1 + 0.5 + 0.25 = (11.75)_{10}$$

Binary → Decimal | $(1110.01)_2$

Position :	3	2	1	0	.	–1	–2
Binary :	1	1	1	0	.	0	1

Value:

$1\times(2)^3 + 1\times(2)^2 + 1\times(2)^1 + 0\times(2)^0 + 0\times(2)^{-1} + 1\times(2)^{-2}$

$$8 + 4 + 2 + 0 + 0 + 0.25 = (14.25)_{10}$$

Binary → Decimal | $(1101)_2$

Position :	3	2	1	0
Binary :	1	1	0	1

Value:

$$1\times(2)^3 + 1\times(2)^2 + 0\times(2)^1 + 1\times(2)^0$$

$$8 + 4 + 0 + 1 = (13)_{10}$$

Binary → Decimal | $(10110)_2$

Position :	4	3	2	1	0
Binary :	1	0	1	1	0

Value:

$1\times(2)^4 + 0\times(2)^3 + 1\times(2)^2 + 1\times(2)^1 + 0\times(2)^0$

$$16 + 0 + 4 + 2 + 0 = (22)_{10}$$

Octal → Decimal | $(567.4)_8$

Value:

$5\times(8)^2$ + $6\times(8)^1$ + $7\times(8)^0$ + $4\times(8)^{-1}$

$$320 + 48 + 7 + 0.5 = (375.5)_{10}$$

Octal → Decimal | $(144.2)_8$

Value:

$1\times(8)^2 + 4\times(8)^1 + 4\times(8)^0 + 2\times(8)^{-1}$

$$64 + 32 + 4 + 0.25 = (100.25)_{10}$$

Octal → Decimal | $(31.6)_8$

Value:

$3\times(8)^1$ + $1\times(8)^0$ + $6\times(8)^{-1}$

$$24 + 1 + 0.75 = (25.75)_{10}$$

Hexadecimal → Decimal | $(177.A)_{16}$

Value:

$1\times(16)^2 + 7\times(16)^1 + 7\times(16)^0 + 10\times(16)^{-1}$

$$256 + 112 + 7 + 0.625 = (375.625)_{10}$$

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Hexadecimal → Decimal | $(FF.8)_{16}$

Value:

$15\times(16)^{1}$ + $15\times(16)^{0}$ + $8\times(16)^{-1}$

$$240 + 15 + 0.5 = (255.5)_{10}$$

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15

Hexadecimal → Decimal | $(2D.4)_{16}$

Value:

$2\times(16)^{1} + 13\times(16)^{0} + 4\times(16)^{-1}$

$$32 + 13 + 0.25 = (45.25)_{10}$$

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15